DATA 211 – Introduction to Data Science

Topic 5: Exploratory Data Analysis

Question bank: 40 multiple choice + 10 programming

Questions from DATA 211 / ISE 291 old exams and quizzes (AI_Work). **[new]** marks new questions written in the same style.

Part A: Multiple choice

Q1.

Quiz 6 (T212)

Exploratory Analysis _____.

(A) identifies the best course of action for a given situation

(B) summarizes data

(C) finds unknown relationships and patterns

(D) identifies how change in one variable leads to the changes in another variable

Answer: (C). Exploratory analysis looks for unknown relationships and patterns (A is prescriptive, B descriptive, D mechanistic).

My answer

Q2.

Quiz 3 (T253, Version C)

Descriptive analysis typically _____.

(A) Facilitates analyzing and summarizing the data

(B) Involves recognizing patterns and detecting anomalies/unusual events

(C) Includes making predictions

(D) Deciding the best course of action

Answer: (A). Descriptive analysis analyzes and summarizes the data (B is diagnostic, C predictive, D prescriptive).

My answer

Q3.

Practice Questions Topic 6

Which of the following is not involved in the descriptive data analysis?

(A) Measures of centrality

(B) Dispersion of a distribution

(C) Data visualization

(D) Relationship between variables

Answer: (D). Descriptive analysis uses plots, centrality and dispersion; relationships between variables are studied by correlation (diagnostic) or exploratory analysis.

My answer

Q4. Practice Questions Topic 6

Suppose there is a significant drop in sales of a company in a specific region during a particular month. To find why this happened, the company conducts a causal analysis. What kind of data analysis/analytics will be conducted?

(A) Prescriptive Analysis

(B) Regression Analysis

(C) Diagnostic Analytics

(D) Predictive Analytics

Answer: (C). "Why did something happen?" (causal analysis) is diagnostic analytics.

My answer

Q5. Quiz 3 (T253, Sec 03, Version B)

Diagnostic analysis typically _____.

(A) Facilitates analyzing and summarizing the data

(B) Involves recognizing patterns and detecting anomalies/unusual events

(C) Includes making predictions

(D) Deciding the best course of action

Answer: (B). Diagnostic techniques recognize patterns, detect anomalies and surface unusual events.

My answer

Q6. Final (T213, solved)

Correlation is an appropriate tool for:

(A) Descriptive Analysis

(B) Diagnostic Analytics

(C) Mechanistic Analysis

(D) Prescriptive Analysis

Answer: (B). Correlation is one of the most frequently used techniques in diagnostic analytics.

My answer

Q7. Practice Questions Topic 6

Which of the following is not involved in the predictive data analysis?

(A) Collect and clean the data

(B) Make predictions

(C) Identify the patterns

(D) Find reason why something happened

Answer: (D). Predictive analytics = collect, clean, identify patterns, make predictions; finding the reason is diagnostic.

My answer

Q8. Quiz 3 key (T253, Dr. Jamal)

Which of the following is involved in the predictive data analysis?

(A) Finding all outliers

(B) Identifying patterns

(C) Finding how change in one variable leads to a change in another variable

(D) Finding reasons why something has happened

Answer: (B). Identifying patterns is step 3 of predictive analytics (C is mechanistic, D diagnostic).

My answer

Q9. Final (T241, Code A)

The regression technique is one of the most frequently used techniques in the following categorization of analysis:

(A) Mechanistic Analysis

(B) Descriptive Analysis

(C) Exploratory Analysis

(D) PCA Analysis

Answer: (A). Regression is one of the most frequently used techniques in mechanistic analysis.

My answer

Q10. Practice set P1

Which of the following is/are included in Mechanistic Analysis?

I. Regression
II. Collecting data
III. Cleaning data
IV. Correlation

(A) I and II only
(B) III and IV only
(C) I, II, III and IV
(D) I only

Answer: (D). Mechanistic analysis uses regression; collecting/cleaning are predictive-analytics steps and correlation is diagnostic.

My answer

Q11. Practice set P7

_____ describes strength and direction of the relationship between two variables.

(A) Regression
(B) PCA
(C) Correlation
(D) Classification

Answer: (C). Correlation has two properties: strength and direction.

My answer

Q12. Practice set P6

Strength of the correlation indicates

(A) how one variable would change its value as the value of the other variable changes
(B) how closely two variables are related to each other
(C) positive/negative relationship
(D) no relationship

Answer: (B). Strength = how closely the two variables are related; A and C describe direction.

My answer

Q13.

Practice set P6

If the values of two variables move in the same direction,

(A) The correlation is said to be linear

(B) The correlation is said to be non-linear

(C) The correlation is said to be negative

(D) The correlation is said to be positive

Answer: (D). If one increases (decreases) and the other increases (decreases), the correlation is positive.

My answer

Q14.

Quiz 6 (T212)

If the value of correlation is positive, then the points in a scatter diagram tend to cluster

(A) From lower left corner to upper right corner

(B) From lower right corner to upper right corner

(C) From lower right corner to upper left corner

(D) From lower left corner to lower right corner

Answer: (A). Positive correlation: y increases as x increases, so points rise from lower left to upper right.

My answer

Q15.

Practice Questions Topic 6

Which one is true about correlation?

(A) Correlation can be any number

(B) It can be from -1 to 1

(C) Correlation cannot be zero

(D) Strongest correlation is 10

Answer: (B). Pearson's r satisfies $-1 \leq r \leq 1$.

My answer

Q16. Quiz 3 (T253, Version C)

Which of the following statements regarding Pearson's correlation analysis is **incorrect**?

(A) A high, statistically significant correlation coefficient between two variables is insufficient evidence to establish a directional, causal path

(B) A correlation coefficient of 0 guarantees that there is no nonlinear relationship

(C) Pearson's correlation coefficient is invariant to linear scaling, meaning its value remains identical if a variable is multiplied by a positive constant

(D) The value of the Pearson correlation coefficient is bounded strictly between -1 and 1, inclusive

Answer: (B). $r = 0$ only means no *linear* correlation; a strong nonlinear relationship can still exist.

My answer

Q17. Quiz 6 (T212)

Which value represents a very weak correlation?

(A) -0.9

(B) 0.5

(C) 0.1

(D) $+9.6$

Answer: (C). 0.1 is close to 0 (essentially no correlation); -0.9 is very strong, 0.5 is some correlation, 9.6 is impossible.

My answer

Q18. Practice set P5

Which of the following is the strongest correlation score?

(A) 0.1

(B) 0.80

(C) -0.85

(D) 0.83

Answer: (C). Strength is the distance from 0: $|-0.85| = 0.85$ is the largest.

My answer

Q19. Final 291 (Code 002)

A data scientist calculated the correlation between two financial variables as -0.945. Therefore, the conclusion should be:

(A) The two variables are weakly correlated

(B) The two variables are very strongly correlated

(C) The two variables are positively correlated

(D) Error in calculation, since correlation value cannot be negative

Answer: (B). $|r| = 0.945 > 0.8$: very strong (negative) correlation.

My answer

Q20. Final 291 (Code 002)

Which of the following can be true about the correlation between any two numerical variables, x and y?

(A) It depicts how variable x is related to variable y

(B) Positive correlation indicates strong correlation between x and y

(C) Negative correlation indicates that x decreases as y decreases

(D) Negative correlation indicates that x increases as y decreases

Answer: (D). Negative direction: if one increases the other decreases. Sign does not give strength (B), and correlation does not depict how they are related (A).

My answer

Q21. Practice set P4

How to calculate the Pearson's correlation with Python?

(A) `df.corr(method='pearson')`

(B) `df.correlation('pearson')`

(C) `np.corr(method='pearson')`

(D) `pd.corr(method='pearson')`

Answer: (A). `DataFrame.corr()` computes pair-wise Pearson correlation (Pearson is the default method).

My answer

Q22. Practice set P7

Given a dataframe `df` of 4 columns (`x1`, `x2`, `x3`, `x4`), where `x4` is categorical and all other variables are

numerical, which of the following code will produce the pair-wise correlation between the numerical variables?

(A) `df.drop('x4',axis=0).corr()`

(B) `df['x1','x2','x3'].corr()`

(C) `df.drop('x4',axis=1).corr()`

(D) A & B

Answer: (C). A column is dropped with `axis=1`; A and B both raise a `KeyError` (B needs double brackets).

My answer

Q23.

Quiz 3 key (T253, Dr. Jamal)

The following commands are used to show (`corr` is the correlation matrix of `df` and `y` is the output column):

```
corr=corr.apply(lambda x: np.abs(x))
sorted_corr = corr.sort_values(by=['y'], ascending=False)
sorted_corr = sorted_corr['y'].index
print('Results: ',sorted_corr[1:6].tolist())
```

(A) 6 input variables which have least relation with output variable

(B) relationship between the input and output variables

(C) top 5 input variables related to output variable

(D) top 5 principal components

Answer: (C). Columns are sorted by $|r|$ with `y` in descending order; index 0 is `y` itself, so `[1:6]` gives the 5 most correlated inputs.

My answer

Q24.

Final (T241, Code A)

The purpose of principal component analysis (PCA) is the following:

(A) Data transformation

(B) Data cube aggregation

(C) Dimensionality reduction

(D) Feature construction

Answer: (C). PCA is a dimensionality reduction technique, useful for visualization and exploration.

My answer

Q25. Practice set P5

Which of the following sentences is False about the principal component analysis?

(A) It is a supervised learning method.

(B) It is used to reduce the number of attributes.

(C) It projects the input data into a lower dimensional linear space.

(D) It is used for data visualization and exploration.

Answer: (A). PCA uses no output/label, so it is unsupervised.

My answer

Q26. Final 291 (Code 002)

The Principal Component Analysis (PCA) projects data into a lower dimension by:

(A) Maximizing the sum of squared errors of projected data

(B) Minimizing the sum of squared errors of the projected data

(C) Maximizing the variance along the projection line

(D) Minimizing the variance along the projection line

Answer: (C). PCA is an orthogonal projection that maximizes the variance of the projected data.

My answer

Q27. Practice Questions Topic 6

Assume that we have data that contains five numerical columns. We performed Principal Component Analysis (PCA) on these columns and computed three PCs; which of the following is TRUE?

(A) All PCs are linear combinations of all the numerical columns

(B) Only the second PC is a linear combination of the numerical columns

(C) Only the first PC is a linear combination of the numerical columns

(D) Only the third PC is a linear combination of the numerical columns

Answer: (A). Every principal component is a linear combination of the original (input) columns.

My answer

Q28. Practice set P1

Which of the following statements is/are TRUE about PCA?

I. All the principal components are ranked randomly regardless of how much variation they can capture.

II. It projects data onto a lower dimensional linear space.

III. The FIRST principal component gives the maximum variation on the principal direction (which is a linear combination of the original columns).

IV. All principal components are orthogonal to each other.

(A) III and IV only

(B) I, II, III and IV

(C) II, III and IV only

(D) I and II only

Answer: (C). PCs are ranked by the variation they capture, so I is false; II, III and IV are true.

My answer

Q29. Practice set P8

In PCA with 3 principal components, which of the following is NOT TRUE?

(A) PC2 is orthogonal to PC1

(B) The variance captured by PC2 is the minimum

(C) The variance captured by PC1 is the maximum

(D) The principal components are linear combinations of some/all the input variables

Answer: (B). With 3 PCs, PC3 captures the least variance, not PC2.

My answer

Q30. Quiz 6 (T212)

If we project data using PCA, which cannot be the variances captured by the first two PCs?

(A) 50%, 50%

(B) 70%, 20%

(C) 40%, 50%

(D) 40%, 30%

Answer: (C). PC1 always captures at least as much variance as PC2, so 40% then 50% is impossible.

My answer

Q31. Final 291 (Code 002)

Which of the following can be true about the Principal Components?

(A) The first principal component is orthogonal to the second component, but not to the other components, if any

(B) It is possible that the first principal component captures less than 50% of the variation in the data

(C) It is possible that the second principal component captures more variation than the first principal component

(D) Each principal component corresponds to only one of the original variables

Answer: (B). PC1 has the largest share, but with several similar columns that share can be below 50%. All PCs are mutually orthogonal (A false), ranked (C false) and are combinations of columns (D false).

My answer

Q32.

Final (T213, solved)

The number of principal components (PCs) should be equal to:

(A) number of samples (or records)

(B) number of dimensions (or columns)

(C) the average of the number of samples (or records) and the number of dimensions (or columns)

(D) number of samples (or records) or number of dimensions (or columns), whichever is smaller

Answer: (D). Number of PCs = min(records, columns).

My answer

Q33.

Final (T241, Code A)

Consider a dataframe of numerical variables with shape $(20, 7)$; then a possible number of principal components is ____.

(A) 21

(B) 19

(C) 20

(D) 6

Answer: (D). At most $\min(20, 7) = 7$ PCs can be computed, so only 6 is possible.

My answer

Q34.

Practice set P8

If you have data in which all columns are non-numerical, and you want to do Principal Component Analysis (PCA), what is the best option?

(A) Label or One-hot Encoding

(B) Converting the columns to numeric

(C) Dropping the columns

(D) A & B

Answer: (D). Only numerical columns can be used, so text columns must be encoded/converted to numbers first.

My answer

Q35.

Final 291 (Code 002)

When obtaining the Principal Components from a given data, we should always:

(A) Exclude the output variable for a valid analysis on the relationship between input and output variables.

(B) Include the output variable for the PCA to perform a better separation accuracy of input and output relationship.

(C) Include all categorical columns using label encoder.

(D) Include all the numerical columns (either input or output).

Answer: (A). An OUTPUT column must not be used as a PCA input (e.g. `economy` is dropped in the case study).

My answer

Q36.

Quiz 3 key (T253, Dr. Jamal)

Which of the following is False about the principal component analysis?

(A) The first PC has the max variance

(B) All PCs are linear combinations of the considered numerical columns

(C) The sum of explained variances (ratios) of all PCs is equal to 1

(D) The PCs have the same units as the original columns

Answer: (D). A PC mixes several columns (e.g. weight, displacement, mpg), so it has no single original unit.

My answer

Q37.

Practice set P3

Which of the following statements can be added to the code below to perform PCA on all numerical variables of the DataFrame `df` excluding the `'output'` variable?

```
import numpy as np
import pandas as pd
df = pd.read_csv('dataset.csv', delimiter = ',')
num_cols = df.select_dtypes(exclude='object').columns
pca_input_cols = num_cols.drop(['output'])
from sklearn.decomposition import PCA
pca = PCA(n_components=3)
```

(A) `p = pca.fit_transform(df[pca_input_cols])`

(B) `p = pca.fit_transform(num_cols)`

(C) `p = pca.fit(num_cols)`

(D) `p = pca.fit_transform(df[num_cols])`

Answer: (A). Fit on the data of the input columns only; B and C pass column names, D still includes `output`.

My answer

Q38.

Quiz 3 (T253, Sec 03, Version B)

The DataFrame `df` has only the following columns (392 rows):

```
model_year      392 non-null   int64
cylinders       392 non-null   int64
displacement    392 non-null   float64
horsepower      392 non-null   int64
weight          392 non-null   int64
acceleration    392 non-null   float64
mpg             392 non-null   float64
economy         392 non-null   int64
```

Based on these details, which of the following code snippets correctly performs Principal Component Analysis (PCA)?

(A) `pca = PCA(n_components=13); principalComponents = pca.fit_transform(df)`

(B) `pca = PCA(n_components=4); principalComponents = pca.fit_transform(df)`

(C) `pca = PCA(n_components=9); principalComponents = pca.fit_transform(df)`

(D) `pca = PCA(n_components=10); principalComponents = pca.fit_transform(df)`

Answer: (B). With 8 columns at most $\min(392, 8) = 8$ PCs exist; 13, 9 and 10 raise a `ValueError`.

My answer

Q39.

Practice set P3

Select the correct code to show how much variation is captured by each of the principal components for the given DataFrame `df`.

```
from sklearn.decomposition import PCA
pca = PCA(n_components=3)
```

(A) `pca.components_`

(B) `df.explained_variance_ratio_`

(C) `pca.explained_variance_ratio_`

(D) `pca.fit_transform`

Answer: (C). The variance captured by each PC is in `pca.explained_variance_ratio_`; `components_` holds the coefficients.

My answer

Q40.

Final (T241, Code A)

Given `df` is a dataframe with all numerical variables and PCA is imported from `sklearn.decomposition`:

```
pca=PCA(n_components=2)
principal_components=pca.fit_transform(df)
print(pca.explained_variance_ratio_)
```

Which output is correct for the above code?

(A) `[0.8, 0.15]`

(B) `[0.5, 0.3, 0.2]`

(C) `[0.15, 0.75]`

(D) `[0.8, 0.4]`

Answer: (A). Two values (2 components), in decreasing order, with sum ≤ 1. B has 3 values, C is increasing, D sums to 1.2.

My answer

Part B: Programming

Q41.

Final 291 (Code 002), Problem A-1

Using the dataframe `dfA` below, find the first 3 principal components using all numerical columns. Add the principal component values to the dataframe as `pc1`, `pc2` and `pc3` columns, respectively.

```
dfA = pd.DataFrame({
    'x1': [2.5, 0.5, 2.2, 1.9, 3.1, 2.3, 2.0, 1.0, 1.5, 1.1, 2.8, 0.7],
    'x2': [2.4, 0.7, 2.9, 2.2, 3.0, 2.7, 1.6, 1.1, 1.6, 0.9, 2.6, 0.8],
    'x3': [10, 4, 11, 9, 12, 10, 8, 5, 6, 4, 11, 3],
    'x4': [0.3, 1.2, 0.4, 0.6, 0.2, 0.5, 0.8, 1.1, 0.9, 1.3, 0.3, 1.4],
    'x14': ['yes','no','yes','yes','yes','yes','no','no','no','no','yes','no']})
```

Answer:

```
import pandas as pd
from sklearn.decomposition import PCA

num_cols = dfA.select_dtypes(exclude='object').columns   # x1..x4 (x14 is text)
pca = PCA(n_components=3)
pcs = pca.fit_transform(dfA[num_cols])
dfA['pc1'] = pcs[:, 0]
dfA['pc2'] = pcs[:, 1]
dfA['pc3'] = pcs[:, 2]
display(dfA)
```

My answer

Q42.

Final 291 (Code 002), Problem A-2

Continuing the previous question, construct a scatter plot using the first two principal components of the data. Differentiate the plot using column `x14`. Can the first two components provide a good separation for the data? Explain.

Answer:

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(6, 5))
sns.scatterplot(x='pc1', y='pc2', hue='x14', data=dfA)
plt.title('First two principal components')
plt.show()
print(dfA.groupby('x14')['pc1'].agg(['min', 'max']))
# Yes: all 'yes' points have pc1 > 1.2 and all 'no' points have pc1 < 0.3,
```

```
# so the two groups are clearly separated along pc1 (pc2 adds little).
```

My answer

Q43. Homework 02 (T213), A-5

For the student data below, identify the pair-wise correlation between all the numerical variables. From the results, identify the effect of the `Absences` on the `Total score`.

```
students = {
 'Python background': ['Excellent','Good','Fair','Good','Excellent','Fair','Good','Fair','
    Excellent','Good'],
 'Statistical background': ['Yes','No','No','Yes','Yes','No','Yes','No','Yes','No'],
 'Weekly studying hours': [12, 6, 4, 9, 14, 3, 10, 5, 13, 7],
 'Previous exams': [8, 3, 2, 6, 9, 1, 7, 2, 8, 4],
 'Absences': [1, 5, 7, 2, 0, 8, 2, 6, 1, 4],
 'Class size': [30, 35, 40, 30, 25, 40, 30, 35, 25, 35],
 'Project hours': [40, 22, 15, 35, 45, 12, 38, 18, 42, 25],
 'Total score': [91, 68, 55, 82, 95, 50, 85, 60, 93, 70]}
```

Answer:

```
import pandas as pd
df = pd.DataFrame(students)
corr = df.select_dtypes(include='number').corr()    # Pearson by default
display(corr.round(2))
r = corr.loc['Absences', 'Total score']
print('r(Absences, Total score) =', round(r, 3))    # about -0.995
# Very strong negative correlation: students with more absences
# tend to get a lower total score.
```

My answer

Q44. Homework 02 (T213), A-6

Using the same `students` data, perform a Principal Components Analysis with the corresponding Python library for the numerical columns, and then: (i) print the variances captured by the first four components; (ii) find and print the pair-wise correlation of the four principal components and comment on the values; (iii) construct a scatter plot of the first two principal components, differentiating the points using the `Python background` and `Statistical background` columns.

Answer:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.decomposition import PCA

df = pd.DataFrame(students)
num_cols = df.select_dtypes(include='number').columns
pca = PCA(n_components=4)
pcs = pca.fit_transform(df[num_cols])

# (i)
for i, v in enumerate(pca.explained_variance_ratio_):
    print(f'PC{i+1}: {v*100:0.2f}%')

# (ii) PCs are orthogonal -> their pair-wise correlations are 0
pc_df = pd.DataFrame(pcs, columns=['PC1', 'PC2', 'PC3', 'PC4'])
display(pc_df.corr().round(4))

# (iii)
pc_df['Python background'] = df['Python background']
pc_df['Statistical background'] = df['Statistical background']
sns.scatterplot(x='PC1', y='PC2', hue='Python background',
                style='Statistical background', data=pc_df)
plt.show()
```

My answer

Q45. HW2 (T253) solutions, A-7

For the company data below, compute the pair-wise Pearson correlation matrix of all numerical columns and visualize it as an annotated heatmap (two decimals, a diverging colour map).

```
import numpy as np
companies = {
 'Company Age': [12, 30, 8, 45, 22, 15, 5, np.nan, 27, 18],
 'Employees': [120, 900, 60, 2500, 400, 210, 35, 150, 700, 260],
 'Revenue': [3.1, 40.2, 1.5, 120.0, 12.5, 6.0, 0.8, 4.2, 25.0, 7.5],
 'R&D Spend': [0.4, 3.5, 0.3, 9.8, 1.2, 0.9, 0.2, 0.5, 2.6, 1.0],
 'Carbon Emissions': [55, 310, 20, 820, 140, 70, 10, 60, 260, 90],
 'ESG Score': [62, 48, 75, 40, 58, 66, 80, 64, 51, 60],
 'ESG Certified': ['Yes','No','Yes','No','No','Yes','Yes','Yes','No','Yes']}
```

Answer:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.DataFrame(companies)
corr = df.select_dtypes(include='number').corr()    # pair-wise Pearson r
plt.figure(figsize=(7, 6))
sns.heatmap(corr, annot=True, fmt='.2f', cmap='coolwarm', vmin=-1, vmax=1)
plt.title('Pair-wise Pearson correlation')
plt.show()
```

My answer

Q46. HW2 (T253) solutions, A-8

Using the `companies` data: select the numerical columns, drop rows with missing values, standardize them, and extract the first three principal components. Print the variance explained by each component, the three most influential variables (largest absolute coefficient) for each component, and draw a scatter plot of the first two components coloured by `ESG Certified`.

Answer:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

df = pd.DataFrame(companies)
num = df.select_dtypes(include='number').dropna()
X = StandardScaler().fit_transform(num)
pca = PCA(n_components=3)
pcs = pca.fit_transform(X)
print(np.round(pca.explained_variance_ratio_, 4))

load = pd.DataFrame(pca.components_, columns=num.columns,
                    index=['PC1', 'PC2', 'PC3'])
for pc in load.index:
    print(pc, load.loc[pc].abs().nlargest(3).index.tolist())

plot_df = pd.DataFrame(pcs[:, :2], columns=['PC1', 'PC2'], index=num.index)
plot_df['ESG Certified'] = df.loc[num.index, 'ESG Certified']
sns.scatterplot(x='PC1', y='PC2', hue='ESG Certified', data=plot_df)
plt.show()
```

My answer

Q47. Quiz 03 notebook (Quiz_03.ipynb), A.1

Load the Iris data into a DataFrame `df` with columns `sepal length`, `sepal width`, `petal length`, `petal width` and `class` (use `sklearn.datasets.load_iris`). Add the first two principal components to the DataFrame and name the columns `PC1` and `PC2` (`class` is the output and must not be used).

Answer:

```
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.decomposition import PCA

iris = load_iris()
df = pd.DataFrame(iris.data, columns=['sepal length', 'sepal width',
                                      'petal length', 'petal width'])
df['class'] = iris.target

num_columns = df.select_dtypes(exclude='object').columns
pca_input_columns = num_columns.drop(['class'])
pca = PCA(n_components=2)                          # get pca model
principalComponents = pca.fit_transform(df[pca_input_columns])
df['PC1'] = principalComponents[:, 0]
df['PC2'] = principalComponents[:, 1]
display(df)
```

My answer

Q48. Quiz 03 notebook (Quiz_03.ipynb), A.2

Continuing the previous question, how much variation is captured by each of the principal components? Print each as a percentage with two decimals.

Answer:

```
for i, v in enumerate(pca.explained_variance_ratio_):
    print(f'The variance captured by PCA {i+1} is: {v*100: 0.2f}%')
# PC1 about 92.46%, PC2 about 5.31%  (together about 97.8%)
```

My answer

Q49. Quiz 3 notebook (ISE-291-03), Problem 2.1

For the data below (`Scores` is the output), find and print the top three input variables most correlated with `Scores` (use the absolute value of Pearson's r, so negative correlations count too).

```
data = {
 'Scores': [89, 42, 31, 27, 24, 67, 66, 37, 72, 27, 30, 97, 82, 63],
 'Age': [19, 24, 21, 25, 22, 22, 20, 21, 20, 21, 21, 19, 22, 22],
 'Selected': [1, 1, 1, 0, 0, 1, 1, 0, 0, 1, 0, 1, 0, 1],
 'Scholarship': [6588, 6448, 6345, 8056, 7010, 6474, 6519, 8604, 9234, 5955, 9651, 9984,
   7360, 9529],
 'High_Marks': [1068, 462, 341, 270, 264, 804, 726, 444, 792, 270, 330, 1164, 902, 630],
 'Gender': [1, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 1, 0],
 'North': [1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0],
 'South': [0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 1, 0, 0, 0],
 'East': [0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1]}
```

Answer:

```
import numpy as np
import pandas as pd

data_01 = pd.DataFrame(data)
corr = data_01.corr()
corr = corr.apply(lambda x: np.abs(x))
sorted_corr = corr.sort_values(by=['Scores'], ascending=False)
sorted_corr = sorted_corr['Scores'].index
print('The top three correlated input variables are: ', sorted_corr[1:4].tolist())
# ['High_Marks', 'Age', 'South']   (index 0 is Scores itself)
```

My answer

Q50. Quiz 3 notebook (Quiz_3.ipynb)

For the CPU data below (ERP is the output), display the correlation matrix and print the top two input variables most correlated with ERP (negative values should be considered, using the absolute value).

```
cpu = {
 'MYCT': [125, 29, 29, 29, 29, 26, 23, 23, 23, 23, 400, 400, 60, 50],
 'MMIN': [256, 8000, 8000, 8000, 8000, 8000, 16000, 16000, 16000, 32000, 1000, 512, 2000,
   4000],
 'MMAX': [6000, 32000, 32000, 32000, 16000, 32000, 32000, 32000, 64000, 64000, 3000, 3500,
   8000, 16000],
 'CACH': [256, 32, 32, 32, 32, 64, 64, 64, 64, 128, 0, 4, 65, 65],
 'CHMIN': [16, 8, 8, 8, 8, 8, 16, 16, 16, 32, 1, 1, 1, 1],
 'CHMAX': [128, 32, 32, 32, 16, 32, 32, 32, 32, 64, 2, 6, 8, 8],
 'PRP': [198, 269, 220, 172, 132, 318, 367, 489, 636, 1144, 38, 40, 92, 138],
 'ERP': [199, 253, 253, 253, 132, 290, 381, 381, 749, 1238, 23, 24, 70, 117]}
```

Answer:

```
import numpy as np
import pandas as pd

df = pd.DataFrame(cpu)
corr = df.corr()
display(corr.round(3))
corr = corr.apply(lambda x: np.abs(x))
sorted_corr = corr.sort_values(by=['ERP'], ascending=False)
sorted_corr = sorted_corr['ERP'].index
print('The top two correlated input variables are: ', sorted_corr[1:3].tolist())
# ['PRP', 'MMIN']  (r = 0.988 and 0.935)
```

My answer